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Samantha Ajovalasit, PhD

Data Scientist

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PROFILE

Data Scientist with a PhD in Economics, Management and Statistics and a II-level Master's in Big Data Analytics & Social Mining, 5+ years turning questions from economists, court administrators and EU consortia into statistical and machine learning models that reached production. Owns the full cycle in Python, R and SQL: data ingestion and ETL, feature engineering, regression, classification, clustering and time-series forecasting, NLP and self-hosted LLM pipelines, Monte Carlo simulation, Docker and cloud deployment, dashboards and data storytelling for non-technical stakeholders. Work adopted by GRINS consortium partners and published in PLoS ONE and Risks. Domains: financial risk, NLP and text analytics. Italian native, full professional working proficiency in English, French B1. Available immediately.

EXPERIENCES

Quantitative Researcher & Data Scientist, Public Debt and Financial Risk - University of Palermo **Mar 2023 — Mar 2026**
Project GRINS – Growing Resilient, Inclusive and Sustainable *Researcher (RTD-a), Mathematical Methods for Economics and Finance*

- Translated economists' fiscal questions into a stochastic scenario tree and simulation engine for Debt Sustainability Analysis: probabilistic forecasts with 10th to 90th percentile fan charts across 8+ country and scenario studies, published in Risks (2024) and a Bruegel working paper (2025).
- Built as sole developer and maintained WhatDSApp end to end (Python, Dash, Plotly, Docker): validated multi-sheet Excel ingestion, simulation back end and interactive results, deployed on the AMELIA cloud platform and adopted by GRINS consortium partners for debt scenario.
- Co-authored ESG and sovereign debt research published in Economia Italiana (2025).
- Provisioned and administered the group's GPU inference node (6×RTX 4090), 4 GPUs dedicated to local LLMs: open-source models (Qwen, Deepseek) served with vLLM and Ollama as a multi-user Linux service open to about 15 users.
- Built the data layer over FactSet and Refinitiv Eikon APIs (bond data, yield curves, credit spreads) plus IMF, FRED and Eurostat series, feeding every model with repeatable pulls.
- Developed a statistically validated lasso for sparse Gaussian graphical models of S&P 500 sector returns, detecting the COVID regime shift (NUMTA 2026).
- Tenure in Mathematics for Business: held the course (calculus and financial mathematics) in the Business Administration degree.

Data Scientist, Judicial Performance Analytics - University of Palermo **Apr 2022 — Mar 2023**
Project Just Smart – Smart Justice *Research Fellow in Economics*

- Merged civil and labour registers of 9 courts into one dataset of 133,028 proceedings in Python, and modelled case duration with survival analysis (Kaplan–Meier, stratified Cox PH, time-varying effects) to find backlog drivers by matter and court; results fed recommendations to court administration (working paper, 2026).
- Delivered automated R Markdown reports to the Courts of Appeal of Palermo and Caltanissetta: duration quartiles, backlog by matter, Sankey flows of closed cases.

Data Scientist, Media Market Analytics - Ca' Foscari University, Venice **Apr 2021 — Apr 2022**
Project H2020 Grant 101004488 – EUMEPLAT *Research Fellow in Computer Science*

- Analyzed 40 years of European media market data (press, radio, TV, OTT, ICT), findings published as EU H2020 deliverable D1.5.
- Released a public R Shiny dashboard with cross-country choropleth maps and descriptive statistics of the European media market.
- Lectured the module Advanced Applications of Computer Science for Cultural Heritage: R programming and network analysis labs.

PhD in Economics, Management and Statistics University of Messina, XXXIII Cycle (EQF 8) **Oct 2017 — Dec 2020**

- Ran large-scale NLP on Italian Twitter data (sentiment analysis, text classification, topic modelling) to detect vaccine hesitancy and measure the effect of political communication on vaccination propensity; published in PLoS ONE (2021).
- Built the streaming and REST API collection pipeline (Tweepy) for large-scale social-media data on the Italian response to COVID-19, feeding the thesis analyses.
- Thesis: "Disorientation towards routine immunization, COVID-19 anxiety, and propensity to vaccinate. Analysis based on Twitter data in Italy".
- Taught R for data analysis, wrangling and visualisation in Business Statistics and Data Science for Demographic Processes.

Internship RJC Soft s.r.l. - Pisa **Sep 2017 — Nov 2017**

- Built an agent-based simulation of Pisa's urban traffic system in Python and QGIS for the company's R&D team (Project Sprint).

EDUCATION

II Level Master's in Big Data Analytics & Social Mining, *University of Pisa* (EQF 7) Jan 2017 — May 2018
Coursework: data mining and machine learning, SQL databases, Hadoop, Spark and Hive, information retrieval, big data ethics and privacy, data visualisation. Thesis: Traffic Analysis in Urban Environments via GPS Trajectories.

Master's Degree in Economics, *University of Pisa, Scuola Superiore Sant'Anna* (EQF 7) Mar 2013 — Jul 2016
Thesis: Option Pricing with Monte Carlo and Variance Reduction Technique.

Bachelor's Degree in Economics and Finance, *University of Palermo* (EQF 6) Sep 2006 — Nov 2012
Thesis: Loan Transfer and Securitisation.

Technical Skills

Programming & Data Science Stack	Python (Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn), 5+ years in production; R; SQL (schema design, joins, analytical queries). Git/GitHub (branching, code review), Jupyter, VS Code. MATLAB, Stata.
Machine Learning & Statistics	Supervised and unsupervised learning: regression, classification, clustering (k-means, hierarchical), regularised regression (Lasso, Elastic Net), feature engineering, PCA. Model evaluation (cross-validation, F1, MCC), bootstrap, maximum likelihood. Time-series forecasting, survival analysis, panel econometrics. Ensemble methods (Random Forest, Gradient Boosting). Neural networks with PyTorch (coursework).
NLP, LLM & Generative AI	Text classification, sentiment analysis, NER and topic modelling on Italian and multilingual corpora (spaCy, NLTK, Gensim, TF-IDF, Word2Vec). Agentic workflows with Claude Code, tool-using agents. HuggingFace Transformers fine-tuning. Self-hosted LLM serving (vLLM, Ollama; Qwen, Deepseek), prompt engineering, structured outputs, LLM output validation.
Data Engineering & Deployment	End-to-end ETL: REST APIs, web scraping, multi-sheet Excel, FactSet and Refinitiv Eikon. Docker, Linux administration, cloud deployment (AMELIA), multi-GPU server administration, async job orchestration, conda. PostgreSQL/PostGIS, MySQL, MongoDB. Apache Spark, Hadoop and Hive (Master's coursework).
Visualisation & Communication	Production dashboards in Plotly/Dash and R Shiny, ggplot2, fan charts, choropleth maps. Automated R Markdown reporting, data storytelling for non-technical stakeholders, conference talks, teaching. MS Office, LaTeX.
Quantitative Modelling	Monte Carlo simulation with variance reduction, scenario and sensitivity analysis, stochastic optimisation (GAMS/GAMSPy), CVaR and portfolio optimisation, network analysis.

Additional Training

Summer School in Network Econometrics, *Italian Society of Econometrics – Ca' Foscari University of Venice* Jun 2022
Network extraction, finance application of multi-layer network models.

Fundamentals of Digital and Computational Demography, *Max Planck Institute for Demographic Research, Rostock* Dec 2019
Tools to improve demographic and social science analysis, with a focus on interpreting new kinds of big data.

Summer School in Data Science and Epidemic Models, *University of Trento* Jul 2019
Complex data analysis, digital tracking, human mobility, computational methods and social media analytics.

Summer School on Clustering, Data Analysis and Visualization of Complex Data, *University of Catania* May 2018
Clustering, classification and visualisation of large datasets; numerical analysis and cloud-computing platforms.

Selected Publications

2026 *Statistically Validated Lasso for Gaussian Graphical Models in Financial Networks.* NUMTA 06/2026, p. 35. numta.org
2025 *Are bad governments a threat to sovereign defaults? Effects of political risk on debt sustainability.* WP 01/2025. bruegel.org
2025 *How much of a threat to US debt sustainability is Trump's One Big Beautiful Bill Act?* Analysis, Bruegel. bruegel.org
2025 *ESG factors and sovereign debt.* Economia Italiana 46(3), 145–164. economiaitaliana.org
2024 *Debt Sustainability in the Context of Population Ageing: A Risk Management Approach.* Risks 12(12). mdpi.com
2022 *H2020 European Media Platform – EUMEPLAT – Data Clustering Reports.* eumeplat.eu
2021 *Evidence of disorientation towards immunization on online social media.* PLoS ONE. journals.plos.org

Software and Tools Released

WhatDSApp: full-stack decision-support app for sovereign debt scenario analysis. Stochastic Debt Analysis models for financing optimisation, fan charts and efficient frontier, plug-and-play ingestion, data from Refinitiv Eikon and FactSet; deployed on the AMELIA cloud platform, adopted by GRINS consortium partners. ameliadp.grins.it
EUMEPLAT Dashboard: interactive dashboard for the EUMEPLAT project, data clustering reports. eumeplat.shinyapps.io

Hobbies

Cycling, running, bricolage, reading, programming for fun, car simulations. Former football referee.